

An Enhanced Low-Computational-Complexity Predefined-Time Convergent Zeroing Neural Network for Constrained Time-Varying Quadratic Programming With Kinematic Control of Robotic Manipulator

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Abstract—Time-varying quadratic programming (TVQP) problems can be regarded as a challenging issue in a wide range of engineering applications, frequently incorporating equality, inequality, and bound constraints. By integrating a nonlinear complementarity problem (NCP) formulation, zeroing neural networks (ZNNs) can be generalized to solve TVQP problems effectively with such a full set of constraints. However, three issues may limit its computational efficiency and practical applications: 1) the inherent formulation of NCP functions for handling equality/inequality and boundary constraints substantially expands dimensions of coefficient matrices/vectors and solution-space variables; 2) the conventional ZNN (CZNN) framework inevitably necessitates matrix inversion operations for real-time solutions; and 3) insufficient robustness against noise interference compromises solution accuracy in practical implementations. To overcome these issues and promote solution performance, this article develops an enhanced lower dimension NCP low-computational-complexity (LCC) ZNN (ELNCP-LCCZNN) model for solving TVQP problems with time-varying equality, inequality, and variable boundary constraints. An ELNCP function is designed to reduce the matrix/vector coefficients of the model, and the LCCZNN model is utilized to construct a new dynamic model that eliminates the necessity for matrix inversion during solution. Furthermore, a nonlinear activation function is involved to guarantee predefined-time convergence and strengthen robustness against noise. The theoretical properties of the proposed ELNCP-LCCZNN model are validated through numerical simulations and experiments on robotic manipulator kinematic control. The results corroborate the analysis and demonstrate improved computational efficiency, enhanced noise robustness, and practical implementability compared with existing approaches.

Index Terms—Dynamic recurrent neural network (RNN), equality and inequality constraints (EC/IEC), noise perturbation, nonlinear complementarity problem, time-varying quadratic programming (TVQP).

I. INTRODUCTION

DRIVEN by the exponential growth of industrial automation and intelligent manufacturing technologies, robotic manipulators, as essential components, face increasing demands for enhanced operational flexibility and precision [1], [2]. Kinematic control of robotic manipulators, which reflects trajectory tracking in the workspace [3], directly impacts operational efficiency and safety. In complex industrial environments, robotic arm trajectory tracking must not only meet dynamic performance requirements but also comply with various physical and operational constraints, including obstacle avoidance, joint limitations, and speed and acceleration limits, which are typically formulated as multiple constraints. The constrained kinematic control problem often aims to minimize quadratic cost functions related to energy consumption [4], allocated task time [5], and kinematic positioning error [6]. As a result, the trajectory tracking problem is generally formulated as a time-varying quadratic programming (TVQP) problem with equality, inequality, and boundary constraints [7]. Therefore, the study of quadratic programming problems with such constraints is essential for ensuring control performance, reducing energy costs, and improving operational safety.

Despite extensive research efforts, the solution of TVQP problems remains challenging due to three primary factors: the increased dimensionality induced by inequality and boundary constraints, the computational overhead introduced by unavoidable matrix inversions, and inevitable noise that compromises accuracy and stability. To address these issues, this article develops an enhanced lower dimensional nonlinear complementarity problem (ELNCP)-based low-computational-complexity (LCC) zeroing neural networks (ZNNs) for solving TVQP problems, termed ELNCP-LCCZNN. By introducing an ELNCP function, the proposed approach effectively

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mitigates dimensional expansion during constraint transformation. Furthermore, a predefined-time-convergent LCCZNN equipped with an advanced activation function (AF) is employed to ensure enhanced convergence and robustness. The effectiveness of the proposed method is validated through numerical simulations and physical experiments on a UR3(e) robotic manipulator. In addition, Table I presents a systematic comparison with existing RNN-based TVQP solvers, demonstrating that the proposed method achieves superior overall performance across key metrics relevant to TVQP solution. The main contributions of this article are summarized as follows.

- 1) An ELNCP-LCCZNN model is developed to solve the TVQP problem with equality, inequality, and boundary constraints. Relative to existing NCPs-LCCZNN schemes, it reduces coefficient-matrix dimensionality, thereby improving computational efficiency for online control.
- 2) Integrating the advanced AF into ELNCP-LCCZNN guarantees predefined-time convergence. This design strengthens the predefined-time convergence property over existing ELNCP-based RNNs, enhancing suitability for time-sensitive practical computations.
- 3) The ELNCP-LCCZNN model demonstrates robust behavior under diverse noise perturbations and consistently achieves lower residual errors than representative finite-time convergent schemes.
- 4) Simulation and experiments on the UR3/UR3(e) manipulators are conducted in cases without noise and with noise, demonstrating the superiority of the proposed model as compared to other methods for solving TVQP problems.

II. RELATED WORKS

The constraint consists exclusively of either equality constraints (EC) or inequality constraints (IEC) formulations in QP problems. Extensive studies on EC-QP problems have led to rapid advancements and the development of promising methods for obtaining feasible solutions. In contrast, IEC-QP problems are more challenging and are commonly addressed by transforming IECs into ECs, followed by the application of EC-QP solvers.

A representative approach [8] introduces slack variables to convert IECs into ECs. However, this method demands high computational stability and a restricted choice of initial points, as it may produce solutions that do not satisfy the original IECs, thereby falling outside the feasible parameter space. Furthermore, the introduction of slack variables increases computational complexity by raising the problem's dimensionality. To mitigate these issues, Yu et al. [9] proposed a penalty strategy combined with a varying-parameter recurrent neural network (PS-VP-RNN), which employs both inner and outer penalty functions to incorporate penalty terms into problem formulations with EC and IEC, thereby transforming the constrained problems into unconstrained objective functions. In addition, inspired by the NCP [10], a perturbed Fischer–Burmeister (pFB) function was integrated into the Karush–Kuhn–Tucker (KKT) conditions [11],

thereby extending these conditions to include IECs. The pFB function facilitates the conversion of KKT conditions with IECs into ECs. Subsequently, Zhang et al. [12] developed an obstacle-variable parameter dynamic learning network that integrates primitive NCP functions, penalty functions, and slack variables. However, most existing transformation strategies inevitably lead to increased problem dimensionality, particularly in the presence of boundary constraints, which can significantly reduce computational efficiency.

Once transformed into EC-QP formulations, TVQP problems are traditionally solved using numerical optimization methods such as the interior point method [13], gradient projection method [14], and active-set method [15]. While these methods perform well for static optimization, their computational efficiency and real-time performance are often insufficient for time-varying scenarios. To overcome these challenges, recurrent neural networks (RNNs) [16], [17] have been widely investigated due to their parallel computation capability and suitability for real-time hardware implementation. Early RNN-based approaches, including conventional gradient neural networks (CGNNs) [18], have demonstrated effectiveness in solving dynamic optimization problems. However, CGNN models rely solely on gradient descent directions, which can lead to persistent lagging residual errors in time-varying environments. To overcome this issue, gradient differential network (GDN) [19] and a fractional-order extension of the gradient neural framework [20] have been proposed for solving TVQP problems.

More recently, a novel RNN based on a vector-valued function design, known as the conventional ZNN (CZNN) [21], has attracted significant attention for time-varying optimization. One of CZNN's key advantages is its ability to eliminate the hysteresis residual errors observed in CGNN by incorporating an additional time derivative of time-varying information, thereby offering a more accurate and robust solution. However, current analyses indicate that the CZNN dynamics model inevitably involves matrix inversion operations during its solution process [22], [23], which degrades its computational efficiency. In practical applications, such as kinematic control of robotic manipulators, there is a growing demand for more accurate and robust trajectory tracking, and experimental results [24] suggest that the performance of CZNN deteriorates significantly in the presence of disturbances. Consequently, ongoing research is focused on enhancing these aspects of CZNN performance. A critical factor influencing ZNN effectiveness is the choice of AFs. For instance, Zeng et al. [25] introduced symbolic double idempotent AFs to facilitate finite-time convergence, while Jin et al. [6] proposed predefined fixed-time convergent ZNN (PFT-ZNN), equipped with a power piecewise AF that achieves predefined-time convergence. Furthermore, the design of the energy function within the CZNN framework is crucial for determining overall performance. Li et al. [26] and Li et al. [27] enhanced the robustness of their models against disturbances by introducing integral energy functions and consequently proposed the improved integral-enhanced ZNN (IIE-ZNN) and the triple integral-enhanced ZNN (TIE-ZNN), respectively. Meanwhile, Chen et al. [28] developed an LCCZNN model that replaces

the CZNN formulation with a scalar-valued energy function, thereby avoiding explicit matrix inversion.

In spite of advances, existing methods still face challenges in simultaneously achieving low dimensionality, high computational efficiency, predefined-time convergence, and strong robustness for TVQP problems with inequality and boundary constraints. These limitations motivate the development of the ELNCP-LCCZNN framework proposed in this article.

III. PROBLEM FORMULATION

In this work, let us consider the following TVQP problem with equality, inequality, and variable boundary constraints as follows:

$$\underset{x}{\text{minimize}} \quad \mathbf{x}^T(t)\mathbf{\Lambda}(t)\mathbf{x}(t)/2 + \mathbf{f}^T(t)\mathbf{x}(t) \quad (1)$$

$$\text{subject to} \quad \mathbf{A}(t)\mathbf{x}(t) = \mathbf{b}(t) \quad (1a)$$

$$\mathbf{C}(t)\mathbf{x}(t) \leq \mathbf{d}(t) \quad (1b)$$

$$\mathbf{u}(t) \leq \mathbf{Z}(t)\mathbf{x}(t) \leq \mathbf{v}(t) \quad (1c)$$

where vector $\mathbf{x}(t) \in \mathbb{R}^n$ is the control variable of optimal system, and $\mathbf{\Lambda}(t) \in \mathbb{R}^{n \times n}$, $\mathbf{A}(t) \in \mathbb{R}^{l \times n}$, $\mathbf{C}(t) \in \mathbb{R}^{m \times n}$, and $\mathbf{Z}(t) \in \mathbb{R}^{p \times n}$ denote time-varying coefficient matrices, $\mathbf{f}(t) \in \mathbb{R}^n$, $\mathbf{b}(t) \in \mathbb{R}^l$, and $\mathbf{d}(t) \in \mathbb{R}^m$ are time-varying coefficient vectors, $\mathbf{u}(t), \mathbf{v}(t) \in \mathbb{R}^p$ are the lower and upper bounds for unknown variable $\mathbf{x}(t)$. To ensure that the TVQP admits a unique solution during the whole system operation, the matrix $\mathbf{\Lambda}(t)$ is required to be a positive definite matrix, and matrices $\mathbf{A}(t)$, $\mathbf{C}(t)$, and $\mathbf{Z}(t)$ are of full rank.

A. Original NCP for Solving TVQP

To facilitate the construction of the generalized Lagrangian function to solve the TVQP problem, the boundary inequality (1c) is demanded to be reformulated into a standardized linear inequality. This reformulation is subsequently combined with the original TVQP problem's inequality (1b) to generate a new augmented generalized IEC, which is expressed as follows:

$$\mathbf{Q}(t)\mathbf{x}(t) \leq \mathbf{q}(t) \quad (2)$$

where coefficient matrix $\mathbf{Q}(t) = [\mathbf{C}(t), \mathbf{Z}(t), -\mathbf{Z}(t)]^T \in \mathbb{R}^{(m+2p) \times n}$ and coefficient vector $\mathbf{q}(t) = [\mathbf{d}(t), \mathbf{v}(t), -\mathbf{u}(t)]^T \in \mathbb{R}^{m+2p}$.

Then, the generalized Lagrange function is usually defined as follows:

$$\begin{aligned} \mathcal{L}(\mathbf{x}(t), \lambda(t), \eta(t), t) = & \frac{1}{2}\mathbf{x}^T(t)\mathbf{\Lambda}(t)\mathbf{x}(t) + \mathbf{f}^T(t)\mathbf{x}(t) \\ & + \lambda^T(t)(\mathbf{A}(t)\mathbf{x}(t) - \mathbf{b}(t)) \\ & + \eta^T(t)(\mathbf{Q}(t)\mathbf{x}(t) - \mathbf{q}(t)) \end{aligned} \quad (3)$$

where $\lambda(t) \in \mathbb{R}^l$ and $\eta(t) \in \mathbb{R}^{m+2p}$ represent the Lagrange multipliers, which are also unknown and to be determined in the solution process of the TVQP problem.

According to the KKT condition, the partial derivatives of the Lagrange function concerning all unknown variables are

forced to be zero first. In this situation, the following resultant equations and inequalities are further obtained:

$$\begin{cases} \nabla \mathcal{L}(\mathbf{x}(t), \lambda(t), \eta(t), t) = \mathbf{\Lambda}(t)\mathbf{x}(t) + \mathbf{f}(t) \\ \quad + \mathbf{A}^T(t)\lambda(t) + \mathbf{Q}^T(t)\eta(t) = \mathbf{0} \\ \mathbf{A}(t)\mathbf{x}(t) - \mathbf{b}(t) = \mathbf{0} \\ \eta(t) \geq \mathbf{0}, \mathbf{q}(t) - \mathbf{Q}(t)\mathbf{x}(t) \geq \mathbf{0} \\ (\mathbf{q}(t) - \mathbf{Q}(t)\mathbf{x}(t))\eta(t) = \mathbf{0}. \end{cases} \quad (4)$$

To further transform the inequalities into solvable equations, an NCP function is designed that satisfies the following complementarity conditions [10]:

$$\Psi(\iota, \kappa) = 0 \Leftrightarrow \iota \geq \mathbf{0}, \kappa \geq \mathbf{0}, \iota \odot \kappa = \mathbf{0} \quad (5)$$

where $\Psi(\iota, \kappa)$ is a smooth function, and the most popular NCP function is the pFB function [32] as follows:

$$\Psi_{\text{pFB}}^\delta(\iota, \kappa) = \iota + \kappa - \sqrt{\iota \odot \iota + \kappa \odot \kappa + \delta}, \delta \rightarrow 0_+ \quad (6)$$

where δ represents the perturbed term, and the notation “ $\odot$ ” denotes the Hadamard product operation. Therefore, we have the following equations for solving the TVQP problem:

$$\begin{cases} \mathbf{\Lambda}(t)\mathbf{x}(t) + \mathbf{f}(t) + \mathbf{A}^T(t)\lambda(t) + \mathbf{Q}^T(t)\eta(t) = \mathbf{0} \\ \mathbf{A}(t)\mathbf{x}(t) - \mathbf{b}(t) = \mathbf{0} \\ \Psi_{\text{pFB}}^\delta(\mathbf{q}(t) - \mathbf{Q}(t)\mathbf{x}(t), \eta(t)) = \mathbf{0} \end{cases} \quad (7)$$

where $\Psi_{\text{pFB}}^\delta(\cdot, \cdot) : \mathbb{R}^{m+2p} \times \mathbb{R}^{m+2p} \rightarrow \mathbb{R}^{m+2p}$. Equation (4) can be expressed in the form of a matrix equation as follows:

$$\mathbf{g}(\mathbf{y}(t), t) = \mathbf{G}(t)\mathbf{y}(t) - \boldsymbol{\phi}(t) = \mathbf{0} \quad (8)$$

where

$$\begin{aligned} \mathbf{G} &:= \begin{bmatrix} \mathbf{\Lambda}(t) & \mathbf{A}^T(t) & \mathbf{Q}^T(t) \\ \mathbf{A}(t) & \mathbf{0}_{l \times l} & \mathbf{0}_{l \times (m+2p)} \\ -\mathbf{Q}(t) & \mathbf{0}_{(m+2p) \times l} & \mathbf{I}_{(m+2p) \times (m+2p)} \end{bmatrix}, \\ \boldsymbol{\phi} &:= \begin{bmatrix} -\mathbf{f}(t) \\ \mathbf{b}(t) \\ -\mathbf{q}(t) + \boldsymbol{\varpi} \end{bmatrix}, \quad \mathbf{y} := \begin{bmatrix} \mathbf{x}(t) \\ \lambda(t) \\ \eta(t) \end{bmatrix} \in \mathbb{R}^{n+l+m+2p} \end{aligned}$$

with $\boldsymbol{\varpi} = \sqrt{\boldsymbol{\varrho}(t) \odot \boldsymbol{\varrho}(t) + \boldsymbol{\eta}(t) \odot \boldsymbol{\eta}(t) + \delta}$, $\boldsymbol{\varrho}(t) = \mathbf{q}(t) - \mathbf{Q}(t)\mathbf{x}(t)$. The optimal solution of TVQP (1) with equality, inequality, and variable boundary constraints is transformed into the solution of (8).

B. ELNCP for TVQP

Inspired by lower dimensional NCP (LNCP) [7], an additional decision variable $\chi \in \mathbb{R}^p$ is defined to represent boundary inequality (1c) in the TVQP problem (1) to construct an enhanced LNCP (ELNCP) to enhance the solution

$$\mathbf{Z}(t)\mathbf{x}(t) = \chi(t), \chi(t) \in \mathcal{C} \quad (9)$$

where set $\mathcal{C} \subseteq \mathbb{R}^p$ is a nonempty, closed, and convex set [33], which has the mathematical form $\mathcal{C} = \{\chi(t) \in \mathbb{R}^p | \mathbf{u}_i(t) \leq \chi_i(t) \leq \mathbf{v}_i(t), i = 1, 2, \dots, p\}$. Based on the definition

above, we can rewrite the optimality condition of problem (1) as follows:

$$\begin{cases} \Lambda(t)x(t) + f(t) + A^T(t)\lambda(t) \\ + C^T(t)\alpha(t) + Z^T(t)\omega(t) = \mathbf{0} \end{cases} \quad (10)$$

$$A(t)x(t) - b(t) = \mathbf{0} \quad (11)$$

$$Z(t)x(t) - \chi(t) = \mathbf{0} \quad (12)$$

$$\Psi_{\text{pFB}}^\delta(d(t) - C(t)x(t), \alpha(t)) = \mathbf{0} \quad (13)$$

$$\chi(t) \in \mathcal{C}, \omega(t) \in N_{\mathcal{C}}(\chi(t)). \quad (14)$$

Let $\lambda(t) \in \mathbb{R}^m$, $\alpha(t) \in \mathbb{R}^l$, and $\omega(t) \in \mathbb{R}^p$ be the Lagrange multipliers associated with the EC $A(t)x(t) = b(t)$, the IEC $C(t)x(t) \leq d(t)$, and the ECs $Z(t)x(t) = \chi(t)$, respectively. In addition, let $N_{\mathcal{C}}(\chi(t))$ be the convex cone of the set $\mathcal{C}$ at the point $\chi(t)$. We assume that the vector $u(t)$ is strictly less than the vector $v(t)$ ($u(t) < v(t)$). Equation (14) can be expressed in a more refined manner as follows:

$$\begin{aligned} u(t) &\leq \chi(t) \leq v(t), \\ \omega^T(t)(\chi(t) - u(t)) &= \omega^T(t)(v(t) - \chi(t)) = \mathbf{0} \end{aligned} \quad (15)$$

with $\omega(t) \geq \mathbf{0}$. This can be further combined into an inequality by considering the minimum of $\chi(t) - u(t) \geq \mathbf{0}$, $v(t) - \chi(t) \geq \mathbf{0}$. In conclusion, nonlinear complementarity constraints are as follows:

$$\begin{aligned} s &= \min(\chi(t) - u(t), v(t) - \chi(t)) \geq \mathbf{0}, \\ \omega(t) &\geq \mathbf{0}, s \odot \omega(t) = \mathbf{0}. \end{aligned} \quad (16)$$

Consequently, (10)–(14) can be formulated as a concise nonlinear equation by integrating (6) into NCP functions while introducing the output of the symbolic coefficients $J = \text{sign}(\chi(t) - (v(t) + u(t))/2)$ in conjunction with s

$$\begin{cases} \Lambda(t)x(t) + f(t) + A^T(t)\lambda(t) \\ + C^T(t)\alpha(t) + JZ^T(t)\omega(t) = \mathbf{0} \\ A(t)x(t) - b(t) = \mathbf{0} \\ \Psi_{\text{pFB}}^\delta(d(t) - C(t)x(t), \alpha(t)) = \mathbf{0} \\ \Psi_{\text{pFB}}^\delta(s, \omega(t)) = \mathbf{0} \end{cases} \quad (17)$$

where the function $\text{sign}(\cdot)$ of J is defined a symbolic function. Rewrite (17) as an equation of the form (8)

$$g(y(t), t) = G(t)y(t) - \phi(t) = \mathbf{0} \quad (18)$$

where

$$G := \begin{bmatrix} \Lambda(t) & A^T(t) & C^T(t) & JZ^T(t) \\ A(t) & \mathbf{0}_{l \times l} & \mathbf{0}_{l \times m} & \mathbf{0}_{l \times p} \\ -C(t) & \mathbf{0}_{m \times l} & \mathbf{I}_{m \times m} & \mathbf{0}_{m \times p} \\ -JZ(t) & \mathbf{0}_{p \times l} & \mathbf{0}_{p \times m} & \mathbf{I}_{p \times p} \end{bmatrix},$$

$$\phi := \begin{bmatrix} -f(t) \\ b(t) \\ -d(t) + \varpi_1 \\ v(t) + \varpi_2 \end{bmatrix}, \quad y := \begin{bmatrix} x(t) \\ \lambda(t) \\ \alpha(t) \\ \omega(t) \end{bmatrix} \in \mathbb{R}^{n+l+m+p}$$

with $v(t) = 1/2(u(t) - v(t) - Ju(t) - Jv(t))$ in vector ϕ , $\varpi_1 = \sqrt{\varrho(t) \odot \varrho(t) + \alpha(t) \odot \alpha(t) + \delta}$, $\varpi_2 = \sqrt{s \odot s + \omega(t) \odot \omega(t) + \delta}$, and $\varrho(t) = d(t) - C(t)x(t)$.

The optimal solution of TVQP (1) with boundary IECs is transformed into solving a lower dimensional (18).

IV. NEURAL NETWORK MODEL DESIGN

This section systematically derives the ELNCP-LCCZNN model, which integrates an ELNCP to formulate the LCCZNN with predefined-time convergence. To demonstrate the superiority of the ELNCP function, we compare three models: NCP-LCCZNN, LNCP-LCCZNN, and ELNCP-LCCZNN. Furthermore, a detailed theoretical analysis of the proposed model is presented, demonstrating that the model exhibits superior convergence performance and robustness.

A. LCCZNN Model

To enable the LCCZNN model to achieve predefined time convergence, we introduced a sign-tri-power exponential AF (STPEAF) [34]

$$\begin{aligned} \Phi(x_i) &= (\kappa_1 \text{sgn}^{1+\frac{1}{2}r_1+\frac{1}{2}r_1 \text{sign}(|\text{sgn}(x_i)|-1)}(x_i) \\ &+ \kappa_2 \text{sgn}^{r_2}(x_i) + \kappa_3 \text{sgn}(x_i)) \exp(|x_i|) \end{aligned} \quad (19)$$

where the parameters κ_1, κ_2 , and $\kappa_3 > 0$ are positive constants; r_1 and r_2 are design parameters, where r_1 is an even integer with $r_1 > 1$, and r_2 is an odd fraction satisfying $0 < r_2 < 1$. The function $\text{sgn}(\cdot)$ is given by

$$\text{sgn}^r(z) = \begin{cases} -|z|^r, & z < 0 \\ 0, & z = 0 \\ |z|^r, & z > 0. \end{cases} \quad (20)$$

With the introduction of the STPEAF, the model exhibits excellent convergence and robustness, as supported by a corresponding theoretical analysis presented in Section IV-B.

Next, we derive specific models for three NCPs-LCCZNN.

1) *ELNCP-LCCZNN Model*: For problem (1), this article develops an ELNCP-LCCZNN framework that improves computational efficiency over CZNN and mitigates the dimensional growth of coefficient matrices and decision variables introduced by NCP-function transformations. To further enhance predefined-time convergence and solution robustness, we employ the STPEAF. In this section, a scalar-valued error function is introduced to construct the LCCZNN model, which is formulated as follows:

$$\varepsilon(t) = \frac{1}{2} \|g(y(t), t)\|_2^2 = \frac{1}{2} g^T(t)g(t). \quad (21)$$

A neural dynamic design formulation is constructed based on the newly defined error function

$$\dot{\varepsilon} = -\gamma \Phi(\varepsilon) \quad (22)$$

where γ is introduced as a learning gain that scales the convergence intensity of the solution dynamics, and $\Phi(\cdot)$ denotes the array of AF, whose elements are scalar mappings defined as monotonically increasing odd functions. Since

$$\dot{\varepsilon} = \frac{\partial \varepsilon}{\partial y} \frac{dy}{dt} + \frac{\partial \varepsilon}{\partial t} = g^T(t) \left(\frac{\partial g}{\partial y} \frac{dy}{dt} + \frac{\partial g}{\partial t} \right) \quad (23)$$

where $\partial g / \partial y = G$, the first-order error-decaying design equation (22) can be reformulated as follows:

$$g^T(t) \left(G \dot{y}(t) + \frac{\partial g}{\partial t} \right) = -\gamma \Phi \left(\frac{\|g(t)\|_2^2}{2} \right). \quad (24)$$

Finally, the LCCZNN model is derived as

$$\mathbf{g}^T(t)G\dot{\mathbf{y}}(t) = -\gamma\Phi\left(\frac{\|\mathbf{g}(t)\|_2^2}{2}\right) - \mathbf{g}^T(t)(\dot{\mathbf{G}}\mathbf{y}(t) - \dot{\boldsymbol{\phi}}) \quad (25)$$

with the coefficient matrix $\mathbf{g}^T(t)G \in \mathbb{R}^{1 \times n}$ and

$$\dot{\boldsymbol{\phi}} = \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \\ \dot{\phi}_3 \\ \dot{\phi}_4 \end{bmatrix} = \begin{bmatrix} -\dot{\mathbf{f}}(t) \\ \dot{\mathbf{b}}(t) \\ -\dot{\mathbf{d}}(t) + \rho_1\dot{\mathbf{q}}(t) + \rho_2\dot{\mathbf{x}}(t) \\ \dot{\mathbf{v}}(t) + \rho_3\dot{\mathbf{s}} + \rho_4\dot{\boldsymbol{\omega}}(t) \end{bmatrix}$$

where $\dot{\mathbf{q}}(t) = \dot{\mathbf{d}}(t) - \dot{\mathbf{C}}(t)\mathbf{x}(t) - \dot{\mathbf{C}}(t)\dot{\mathbf{x}}(t)$, $\rho_1 = \text{diag}(\varrho \oslash \varpi_1)$, $\rho_2 = \text{diag}(\alpha \oslash \varpi_1)$, $\rho_3 = \text{diag}(s \oslash \varpi_2)$, $\rho_4 = \text{diag}(\omega \oslash \varpi_2)$, $\dot{\mathbf{v}}(t) = 1/2(\dot{\mathbf{u}}(t) - \dot{\mathbf{v}}(t) - \mathbf{J}\dot{\mathbf{u}}(t) - \mathbf{J}\dot{\mathbf{v}}(t))$, $\dot{\mathbf{s}} = d(\min(\chi(t) - \mathbf{u}(t), \mathbf{v}(t) - \chi(t)))/dt = 1/2(\mathbf{U} + \mathbf{V} + \mathbf{J}\mathbf{V} - \mathbf{J}\mathbf{U})$, there is $\mathbf{U} = \dot{\mathbf{Z}}(t)\mathbf{x}(t) + \mathbf{Z}(t)\dot{\mathbf{x}}(t) - \dot{\mathbf{u}}(t)$, $\mathbf{V} = \dot{\mathbf{v}}(t) - \dot{\mathbf{Z}}(t)\mathbf{x}(t) - \mathbf{Z}(t)\dot{\mathbf{x}}(t)$. The symbol “ $\oslash$ ” represents the Hadamard division.

Ultimately, the implicit dynamics of the ELNCP-LCCZNN are transformed into explicit dynamics by exploiting the ℓ_2 -norm property of vectors, thereby eliminating the need for matrix inversion. The ELNCP-LCCZNN model is formulated as

$$\dot{\mathbf{y}}(t) = -\frac{\mathcal{F}^T \mathbf{g}}{\|\mathcal{F}^T \mathbf{g}\|_2^2 + \epsilon} \cdot \left(\gamma\Phi\left(\frac{\|\mathbf{g}\|_2^2}{2}\right) + \mathbf{g}^T(\mathcal{H}\mathbf{y} - \boldsymbol{\xi}) \right) \quad (26)$$

where $\epsilon > 0$ denotes a regularization term to avoid singularities, and

$$\mathcal{F} := \begin{bmatrix} \Lambda(t) & A^T(t) & C^T(t) & \mathbf{J}Z^T(t) \\ A(t) & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ (\rho_1 - I)C(t) & \mathbf{0} & I - \rho_2 & \mathbf{0} \\ (\rho_3 - I)JZ(t) & \mathbf{0} & \mathbf{0} & I - \rho_4 \end{bmatrix},$$

$$\mathcal{H} := \begin{bmatrix} \dot{\Lambda}(t) & \dot{A}^T(t) & \dot{C}^T(t) & \mathbf{J}\dot{Z}^T(t) \\ \dot{A}(t) & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ (\rho_1 - I)\dot{C}(t) & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ (\rho_3 - I)J\dot{Z}(t) & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix},$$

$$\boldsymbol{\xi} := \begin{bmatrix} -\dot{\mathbf{f}}(t) \\ \dot{\mathbf{b}}(t) \\ \rho_1\dot{\mathbf{d}}(t) - \dot{\mathbf{d}}(t) \\ \rho_3\dot{\mathbf{v}}(t) + \dot{\mathbf{v}}(t) \end{bmatrix}.$$

Remark: The regularization term $\epsilon > 0$ is introduced in (26) to avoid singularities caused by the vanishing denominator $\|\mathcal{F}^T \mathbf{g}\|_2^2$ during the convergence process. As the system approaches the optimal solution, the residual error $\mathbf{g}(t)$ tends to zero, which can cause computational instability or even divergence due to division by a diminishing quantity. Introducing a small positive regularization value ensures computational stability and continuous dynamics while preserving convergence properties. In practice, the value of ϵ should be small enough to minimize its effect on optimality, yet large enough to maintain numerical stability. Empirically, a good choice is to set ϵ to be 1–3 orders of magnitude smaller than the squared norm of the initial error. This range provides stable convergence without introducing a noticeable bias in the final solution.

The above formulation neglects noise and exogenous disturbances, which are unavoidable in neural computation and

signal generation. To enhance practical applicability, we incorporate an additive noise term into the neural dynamics. The ELNCP-LCCZNN model with noise is expressed as

$$\dot{\mathbf{y}}(t) = \frac{\mathcal{F}^T \mathbf{g}}{\|\mathcal{F}^T \mathbf{g}\|_2^2 + \epsilon} \cdot \left(\gamma\Phi\left(\frac{\|\mathbf{g}\|_2^2}{2}\right) + \mathbf{g}^T(\mathcal{H}\mathbf{y} - \boldsymbol{\xi}) \right) + \mathcal{N} \quad (27)$$

where $\mathcal{N} = \mathbf{g}^T \mathbf{F} \mathbf{n}(t)$, $\mathbf{n}(t) \in \mathbb{R}^n$ denotes a noise vector. Similarly, we present another theorem to elucidate our findings regarding the robustness of the model.

2) *LNCP-LCCZNN Model:* Different from ELNCP-LCCZNN, $\mathcal{F}$, $\mathcal{H}$, $\boldsymbol{\xi}$ of expression (26) are defined in LNCP-LCCZNN model as presented in [7] as follows:

$$\mathcal{F} := \begin{bmatrix} \Lambda(t) & A^T(t) & C^T(t) & Z^T(t) \\ A(t) & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ (\rho_1 - I)C(t) & \mathbf{0} & I - \rho_2 & \mathbf{0} \\ (\rho_3 - I)Z(t) & \mathbf{0} & \mathbf{0} & \rho_7(I - \rho_4) \\ (I - \rho_5)Z(t) & \mathbf{0} & \mathbf{0} & \rho_8(I - \rho_6) \end{bmatrix},$$

$$\mathcal{H} := \begin{bmatrix} \dot{\Lambda}(t) & \dot{A}^T(t) & \dot{C}^T(t) & \dot{Z}^T(t) \\ \dot{A}(t) & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ (\rho_1 - I)\dot{C}(t) & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ (\rho_3 - I)\dot{Z}(t) & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ (I - \rho_5)\dot{Z}(t) & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix},$$

$$\boldsymbol{\xi} := \begin{bmatrix} -\dot{\mathbf{f}}(t) \\ \dot{\mathbf{b}}(t) \\ \rho_1\dot{\mathbf{d}}(t) - \dot{\mathbf{d}}(t) \\ \dot{\mathbf{v}}(t) - \rho_3\dot{\mathbf{v}}(t) \\ \rho_5\dot{\mathbf{u}}(t) - \dot{\mathbf{u}}(t) \end{bmatrix}.$$

Following the ELNCP-LCCZNN guidelines, ρ_1 and ρ_2 are defined accordingly, while the remaining terms are expressed as $\rho_3 = \text{diag}(\mathbf{r}_1 \oslash \mathbf{l}_1)$, $\rho_4 = \text{diag}(\omega_+ \oslash \mathbf{l}_1)$, $\rho_5 = \text{diag}(\mathbf{r}_2 \oslash \mathbf{l}_2)$, $\rho_6 = \text{diag}(\omega_- \oslash \mathbf{l}_2)$, $\rho_7 = \text{sign}(\omega_+) \oslash \omega$, $\rho_8 = \text{sign}(\omega_-) \oslash \omega$. In the above definition, $\mathbf{r}_1 = \mathbf{v} - \mathbf{y}$, $\mathbf{r}_2 = \mathbf{y} - \mathbf{u}$, $\omega_+ = \max^2(\omega, \mathbf{0})/2$, $\omega_- = \min^2(\omega, \mathbf{0})/2$, $\mathbf{l}_1 = \sqrt{\mathbf{r}_1 \oslash \mathbf{r}_1 + \omega_+ \oslash \omega_+ + \delta}$, $\mathbf{l}_2 = \sqrt{\mathbf{r}_2 \oslash \mathbf{r}_2 + \omega_- \oslash \omega_- + \delta}$.

3) *NCP-LCCZNN Model:* For NCP-LCCZNN, $\dot{\boldsymbol{\phi}}$ of expression (25) is denoted as follows:

$$\dot{\boldsymbol{\phi}} = \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \\ \dot{\phi}_3 \end{bmatrix} = \begin{bmatrix} -\dot{\mathbf{f}}(t) \\ \dot{\mathbf{b}}(t) \\ -\dot{\mathbf{q}}(t) + \rho_1\dot{\mathbf{q}}(t) + \rho_2\dot{\mathbf{q}}(t) \end{bmatrix}$$

where $\rho_1 = \text{diag}(\varrho \oslash \varpi_1)$, $\rho_2 = \text{diag}(\alpha \oslash \varpi_1)$, and $\varrho = \mathbf{q}(t) - \mathbf{Q}(t)\mathbf{x}(t)$, $\dot{\mathbf{q}} = \dot{\mathbf{q}}(t) - \dot{\mathbf{Q}}(t)\mathbf{x}(t) - \mathbf{Q}(t)\dot{\mathbf{x}}(t)$. Thus, $\mathcal{F}$, $\mathcal{H}$, $\boldsymbol{\xi}$ of expression (26) are represented in NCP-LCCZNN as

$$\mathcal{F} := \begin{bmatrix} \Lambda(t) & A^T(t) & \mathbf{Q}^T(t) \\ A(t) & \mathbf{0} & \mathbf{0} \\ (\rho_1 - I)\mathbf{Q}(t) & \mathbf{0} & I - \rho_2 \end{bmatrix},$$

$$\mathcal{H} := \begin{bmatrix} \dot{\Lambda}(t) & \dot{A}^T(t) & \dot{\mathbf{Q}}^T(t) \\ \dot{A}(t) & \mathbf{0} & \mathbf{0} \\ (\rho_1 - I)\dot{\mathbf{Q}}(t) & \mathbf{0} & \mathbf{0} \end{bmatrix},$$

$$\boldsymbol{\xi} := \begin{bmatrix} -\dot{\mathbf{f}}(t) \\ \dot{\mathbf{b}}(t) \\ \rho_1\dot{\mathbf{q}}(t) - \dot{\mathbf{q}}(t) \end{bmatrix}.$$

A comparison of the coefficient matrix and variable dimensions for the three models is presented in Table II, where the models proposed in this article exhibit lower dimensionality compared with existing NCPs-LCCZNN models.

TABLE I
COMPARISONS ON VARIOUS INDEXES FOR DIFFERENT REPRESENTATIVE WORKS ON SOLVING TVQP PROBLEMS

Related works	Equality constraint	Inequality constraint	Bound constraint	Robustness to noise	Dimension reduction	Convergence
[29]	✓	×	×	✓	×	Fixed-time
[6]	✓	×	×	×	×	Predefined fixed-time
[27]	✓	×	×	✓	×	Exponential
[7]	✓	×	✓	×	✓	Finite-time
[8]	✓	✓	×	×	×	Finite-time
[11]	✓	✓	×	×	×	Predefined-time
[19]	✓	✓	×	×	×	Finite-time
[26]	✓	✓	×	✓	×	Finite-time
[30]	✓	✓	×	✓	×	Predefined-time
[12]	✓	✓	✓	×	×	Superexponential
[31]	✓	✓	✓	×	×	Finite-time
[9]	✓	✓	✓	×	✓	Finite-time
This paper	✓	✓	✓	✓	✓	Predefined-time

“✓” represents considered key factors in the index.

“×” represents not considered key factors in the index.

TABLE II

DIMENSION OF COEFFICIENT MATRICES AND LAGRANGE MULTIPLIERS EMPLOYED FOR NCP-LCCZNN, LNCP-LCCZNN, AND ELNCP-LCCZNN FOR TVQP PROBLEMS

Model	NCP-LCCZNN	LNCP-LCCZNN	ELNCP-LCCZNN
$\mathcal{F}$	$(n+l+m+2p) \times (n+l+m+2p)$	$(n+l+m+2p) \times (n+l+m+p)$	$(n+l+m+p) \times (n+l+m+p)$
$\mathcal{H}$	$(n+l+m+2p) \times (n+l+m+2p)$	$(n+l+m+2p) \times (n+l+m+p)$	$(n+l+m+p) \times (n+l+m+p)$
ξ	$n+l+m+2p$	$n+l+m+2p$	$n+l+m+p$
λ	l	l	l
η	$m+2p$	-	-
α	-	m	m
ω	-	p	p

B. Theoretical Analyses and Results

This section discusses the predefined-time convergence of the ELNCP-LCCZNN model equipped with STPEAF (19) toward the optimal solution of the TVQP problem, supported by rigorous theoretical guarantees. Furthermore, the analysis demonstrates that the proposed model preserves stable convergence behavior in the presence of additive noise disturbances.

Theorem 1 (Convergence): For problem (1), the STPEAF-assisted ELNCP-LCCZNN dynamics in (26) drive the state to the theoretical solution within the predefined time t_{AF} , as

$$t_{AF} = \frac{1}{\gamma\kappa_1 r_1 \exp(1)} + \frac{1}{\gamma\kappa_2(1-r_2)}. \quad (28)$$

Proof: Note that the predefined convergence time is independent of the sign of the initial error. Therefore, it is sufficient to consider $\varepsilon_{ini}(t) \geq 0$ without loss of generality. Define the Lyapunov function $V_1(\varepsilon) = |\varepsilon|$, which is positive definite and radially unbounded. Differentiating $V_1(\varepsilon)$ along the solution of (19) gives

$$\begin{aligned} \dot{V}_1(\varepsilon) &= \frac{dV_1}{dt} = \frac{d\varepsilon}{dt} \\ &= -\gamma \left((\kappa_1 \operatorname{sgn}^{1+\frac{1}{2}r_1 + \frac{1}{2}r_1 \operatorname{sgn}(|\operatorname{sgn}(\varepsilon)|-1)}(\varepsilon) \right. \\ &\quad \left. + \kappa_2 \operatorname{sgn}^{r_2}(\varepsilon) + \kappa_3 \operatorname{sgn}(\varepsilon)) \exp(|\varepsilon|) \right). \end{aligned} \quad (29)$$

It follows that $\dot{V}_1 \leq 0$ and $\dot{V}_1 = 0$ holds if and only if $\varepsilon = 0$. Therefore, the Lyapunov function $V(\varepsilon)$ meets the requirements in [35, Lemma 1], implying that the proposed dynamics converge globally and asymptotically to the theoretical solution

$x^*(t)$. Building on this result, we further establish predefined-time convergence of the ELNCP-LCCZNN model.

According to (29), the following differential equation can be obtained:

$$dt = d\varepsilon / \dot{V}_1(\varepsilon). \quad (30)$$

Evaluating the above relation over the interval $[0, t_{AF}]$ yields the following integral form:

$$\begin{aligned} \int_0^{t_{AF}} dt &= \int_0^{t_1} dt + \int_{t_1}^{t_{AF}} dt \\ &= \int_{\varepsilon_{ini}}^{\varepsilon(t_1)} \frac{1}{-\gamma \left((\kappa_1 \varepsilon^{1+r_1} + \kappa_2 \varepsilon^{r_2} + \kappa_3 \varepsilon) \exp(\varepsilon) \right)} d\varepsilon \\ &\quad + \int_1^{\varepsilon(T)} \frac{1}{-\gamma \left((\kappa_1 \varepsilon + \kappa_2 \varepsilon^{r_2} + \kappa_3 \varepsilon) \exp(\varepsilon) \right)} d\varepsilon \\ &\leq \frac{1}{\gamma\kappa_1 \exp(1)} \int_1^{\varepsilon_{ini}} \frac{1}{\varepsilon^{1+r_1}} d\varepsilon \\ &\quad + \lim_{\varepsilon(T) \rightarrow 0} \int_{\varepsilon(T)}^1 \frac{1}{\gamma \left((\kappa_1 \varepsilon + \kappa_2 \varepsilon^{r_2} + \kappa_3 \varepsilon) \exp(\varepsilon) \right)} d\varepsilon \\ &\leq \frac{1}{\gamma\kappa_1 \exp(1)} \cdot \frac{1}{r_1} \left(1 - \frac{1}{\varepsilon_{ini}^{r_1}} \right) \\ &\quad + \lim_{\varepsilon(T) \rightarrow 0} \frac{1}{\gamma} \int_{\varepsilon(T)}^1 \frac{1}{\kappa_2 \varepsilon^{r_2} \exp(\varepsilon)} d\varepsilon \\ &\approx \frac{1}{\gamma\kappa_1 r_1 \exp(1)} + \frac{1}{\gamma\kappa_2(1-r_2)}. \end{aligned} \quad (31)$$

In summary, the ELNCP-LCCZNN model (26) reaches the desired solution within t_{AF} eventually produces

$$t_{AF} < \frac{1}{\gamma\kappa_1 r_1 \exp(1)} + \frac{1}{\gamma\kappa_2(1-r_2)}. \quad (32)$$

All of those complete the proof. $\square$

Theorem 2 (Robustness): Assume that the noise-related term $\mathcal{N}(t)$ satisfies $|\mathcal{N}| \leq \bar{\mathcal{N}}$, where $\bar{\mathcal{N}} > 0$ is an unknown constant. Let $\|\varepsilon\|_2$ be the residual error of the noise-perturbed ELNCP-LCCZNN. Then, the steady-state residual admits the bound $W(2\bar{\mathcal{N}}/\gamma)$, with $W(\cdot)$ denoting the Lambert W function [36]. Furthermore, this bound can be tightened by increasing γ .

Proof: Introduce the Lyapunov candidate $V_2(t) = (\Phi(\varepsilon))^p$, $p = 1 + r_1 + r_2$. Considering $\varepsilon(t) \geq 0$, $\Phi(\varepsilon)$ admits the piecewise form given above. For notational convenience, define

$$P(\varepsilon) = \frac{d\Phi(\varepsilon)}{d\varepsilon} = \begin{cases} P_1, & \varepsilon \geq 1 \\ P_2, & 0 \leq \varepsilon < 1 \end{cases} \quad (33)$$

where $P_1 = (\kappa_1(1+r_1+\varepsilon)\varepsilon^{r_1} + \kappa_2(r_2+\varepsilon)\varepsilon^{r_2-1} + \kappa_3(1+\varepsilon))\exp(\varepsilon)$ and $P_2 = ((\kappa_1+\kappa_3)(1+\varepsilon) + \kappa_2(r_2+\varepsilon)\varepsilon^{r_2-1})\exp(\varepsilon)$. It is straightforward to verify that $P(\varepsilon) > 0$ for all $\varepsilon \geq 0$ and $P_1 \geq P_2$.

Taking the time derivative of $V_2(t)$ along the system trajectory and substituting the error dynamics yields

$$\begin{aligned} \dot{V}_2(t) &= p(\Phi(\varepsilon))^{p-1} P \dot{\varepsilon} = -p(\Phi(\varepsilon))^{p-1} P(\gamma\Phi(\varepsilon) - \mathcal{N}) \\ &\leq -p(\Phi(\varepsilon))^{p-1} P(\gamma\Phi(\varepsilon) - \bar{\mathcal{N}}) \end{aligned} \quad (34)$$

where $P_1 = (\kappa_1(1+r_1+\varepsilon)\varepsilon^{r_1} + \kappa_2(r_2+\varepsilon)\varepsilon^{r_2-1} + \kappa_3(1+\varepsilon))\exp(\varepsilon)$.

Let ϑ be defined by $\gamma\Phi(\vartheta) = 2\bar{\mathcal{N}}$, $\dot{V}_2(t) \leq -p\bar{\mathcal{N}}(\Phi(\varepsilon))^{p-1} P_1$ for all $\varepsilon \geq \vartheta$. The $\mathcal{K}_\infty$ class functions $\alpha(|\varepsilon|) = (\varepsilon \exp(\varepsilon))^p$ and $\beta(|\varepsilon|) = (\Phi(\varepsilon))^p$ is defined. It holds for all $\varepsilon \in [0, +\infty)$ that $\alpha(|\varepsilon|) \leq V_2 \leq \beta(|\varepsilon|)$. Therefore, the conditions in [35] are satisfied. For any initial state, there exists a finite time $T = T(\vartheta, \varepsilon_{ini})$ such that

$$\|\varepsilon\|_2 \leq \alpha^{-1}(\beta(\vartheta)) = W\left(\frac{2\bar{\mathcal{N}}}{\gamma}\right). \quad (35)$$

Consequently, under bounded additive noise, the steady-state residual error of the ELNCP-LCCZNN model is explicitly bounded and decreases monotonically with respect to the learning parameter γ . With an appropriately large γ , the residual bound can be reduced to an arbitrarily small level. The proof is thus concluded. $\square$

V. SIMULATION AND EXPERIMENTAL RESULTS

This section presents numerical simulations and experimental studies to validate the proposed model for solving TVQP problems with equality, inequality, and boundary constraints. The results demonstrate the convergence, robustness, and stable computational performance of the ELNCP-LCCZNN model in both benchmark simulations and manipulator control tasks. Comparative results with existing RNN-based models (CZNN [21], GDN [37], PFT-ZNN [6], IIE-ZNN [26], LCCZNN [35], TIE-ZNN [27], and predefined-time convergent and noise-tolerant ZNN (PTC-NT-ZNN) [38]) further demonstrate the enhanced accuracy, computational efficiency, and superior robustness of the proposed model. The parameter

TABLE III
PARAMETER CONFIGURATIONS OF THE COMPARED MODELS IN COMPUTER SIMULATIONS AND PHYSICAL EXPERIMENT

Model	Parameters Configuration (noise-free: $\gamma = 1$, noise: $\gamma = 10$)
CZNN [21]	NA
GDN [37]	$k = 0.5$
PFT-ZNN [6]	$\varpi = \mu = \xi = 5, \alpha = 3, \beta = 1, \sigma = 1$, and $\delta = 3$
IIE-ZNN [26]	NA
LCCZNN [35]	$r = 0.5$
TIE-ZNN [27]	NA
PTC-NT-ZNN [38]	$\kappa = 0.5, \alpha = 0.8, \zeta = 0.5$
ELNCP-LCCZNN	$\kappa_1 = \kappa_2 = \kappa_3 = 5, r_1 = 4, r_2 = 1/5$

“NA” indicates that no additional parameter settings are specified.

settings are summarized in Table III. All simulations are implemented in MATLAB R2024a on a laptop equipped with an Intel Core Ultra 5 125H CPU and 32-GB RAM. Manipulator experiments are conducted on a UR3(e) robot via a TCP/IP interface at a control frequency of 100 Hz.

Let us consider the following TVQP problem with time-varying equality, inequality, and boundary constraints as follows:

$$\begin{aligned} &\text{minimize } (\cos(t)/4 + 1)x_1^2(t) + (\sin(t)/4 + 1)x_2^2(t) \\ &\quad + \sin(t)x_1(t)x_2(t) + \cos(3t)x_1(t) + \sin(3t)x_2(t) \\ &\text{subject to } \cos(4t)x_1(t) + \sin(4t)x_2(t) = \sin(2t) \\ &\quad \sin(4t)x_1(t) + \cos(4t)x_2(t) \leq 4\sin(2t) \\ &\quad -0.5(\sin(t) + 2) \leq x_1(t), x_2(t) \leq 0.5(\cos(t) + 2). \end{aligned} \quad (36)$$

A. Solving TVQP Problem in the Free of Noise

In this section, to find the optimal state solutions $\mathbf{y}(t) = [\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\alpha}, \boldsymbol{\omega}]^T$, the initial state vector $\mathbf{y}(0) = [\mathbf{x}(0), \boldsymbol{\lambda}(0), \boldsymbol{\alpha}(0), \boldsymbol{\omega}(0)]^T$ is randomly generated within the interval $[0, 1]$. The perturbation term δ of the pFB function is set as 10^{-11} .

In the simulation validation, predefined-time-convergent NCPs-LCCZNN models are constructed using NCP formulations of varying dimensions. Fig. 1 compares the synthesized residual errors, state solutions constrained by boundaries, and computational time costs of the NCP-LCCZNN, LNCP-LCCZNN, and ELNCP-LCCZNN models. As illustrated in Fig. 1(a) and (b), the state solutions produced by the NCPs-LCCZNN model remain strictly within the boundary limits (black-dashed lines), where x^- and x^+ denote the lower and upper bounds of the TVQP problem (36), respectively. In Fig. 1(c), the residuals synthesized by the NCP-LCCZNN and LNCP-LCCZNN models show minor fluctuations, while the majority of residuals remain on the order of 10^{-14} . In contrast, the residuals of the proposed ELNCP-LCCZNN model are considerably more stable. Furthermore, Fig. 1(d) reveals that while maintaining the same magnitude of residual errors, the proposed ELNCP-LCCZNN model achieves significantly smaller computational time costs than the other two models. These results collectively validate our theoretical analysis that dimensionality reduction in the coefficient matrices improves computational efficiency without compromising solution accuracy.

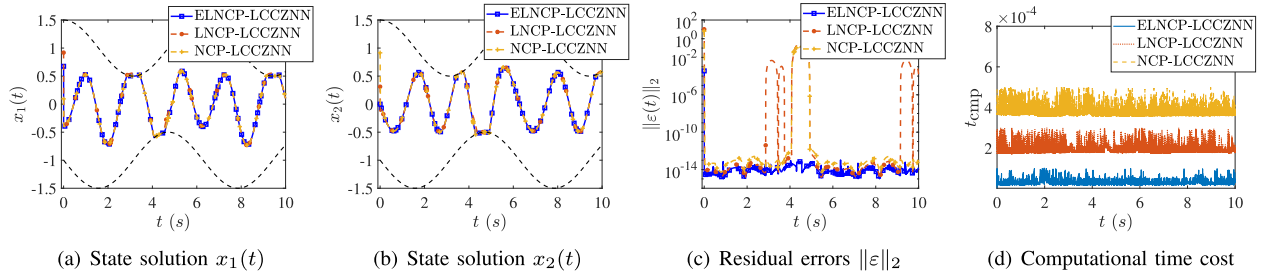


Fig. 1. Computational results obtained from the NCP-LCCZNN, LNCP-LCCZNN, and ELNCP-LCCZNN models when solving the TVQP problem (36). (a) Neural state profiles for x_1 , (b) neural state profiles for x_2 , (c) residual errors generated by the three NCPs-LCCZNN models, and (d) computational time costs of the three NCPs-LCCZNN models with convergence parameter $\gamma = 10$.

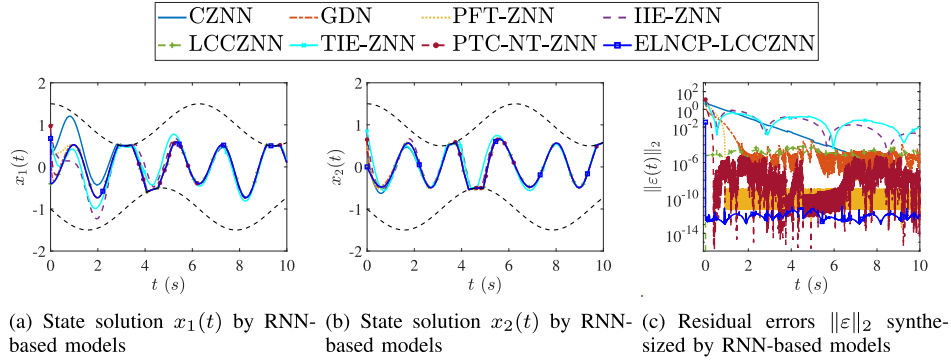


Fig. 2. Computational results obtained from existing RNN-based models, including CZNN, GDN, PFT-ZNN, IIE-ZNN, LCCZNN, TIE-ZNN, and PTC-NT-ZNN when solving the TVQP problem (36). (a) Neural state profiles for x_1 , (b) neural state profiles for x_2 , and (c) residual error profiles produced by the RNN-based methods.

Next, Fig. 2 presents the residual errors of existing RNN-based models, including CZNN, GDN, PFT-ZNN, IIE-ZNN, LCCZNN, TIE-ZNN, and PTC-NT-ZNN when solving the TVQP problem (36). The comparison results in Fig. 2(a)–(c) indicate that, in the absence of noise interference, all models satisfy the boundary constraints and converge steadily to the theoretical solution, validating the effectiveness of existing RNN-based models in solving TVQP problems. Furthermore, performance analysis considering convergence speed and solution accuracy reveals that the proposed model outperforms other methods in both convergence time and solution precision. Specifically, convergence is achieved in approximately 0.001 s, while solution accuracy is maintained at the 10^{-12} order of magnitude, confirming its superior efficiency and precision. These results collectively validate the ELNCP-LCCZNN model's exceptional performance in solving noise-free TVQP problems, offering improvements in both convergence rate and computational accuracy.

B. Solving TVQP Problem Under Noise

In Section V-A, we established that the ELNCP-LCCZNN model exhibits superior solution performance compared to other RNN-based models. Furthermore, compared with existing RNN-based models, the proposed ELNCP-LCCZNN demonstrates remarkable convergence properties under noise-free conditions. However, robustness is a critical consideration in practical applications. To evaluate the robustness of the model against noise robustness, three disturbance profiles are injected into the dynamics: 1) constant noise $\mathbf{n}_1(t) = 5$; 2)

linear time-varying noise $\mathbf{n}_2(t) = 0.5t$; and 3) quadratic time-varying noise $\mathbf{n}_3(t) = 0.5(t - 5)^2$. Unless otherwise stated, the remaining parameters follow Table III. In particular, the learning gain γ is chosen as 10 to moderate the convergence rate.

Consistent with the evaluation methodology in Section V-A, we evaluate the robustness of the models based on convergence speed and solution accuracy, with the results presented in Fig. 3. This figure illustrates the state trajectories and corresponding residual error curves of the RNN-based methods under various noise profiles. The results indicate that the state solutions generated by all the aforementioned RNN-based models continue to satisfy the boundary constraints. However, the residual errors synthesized by CZNN, GDN, and PFT-ZNN models exhibit large deviations under noise interference, which increase over time and fail to decrease further. Furthermore, the IIE-ZNN and TIE-ZNN models also converge gradually to the theoretical solution under noise, albeit with relatively slower convergence rates. The remaining models exhibit varying degrees of fluctuation throughout the solution process as a result of noise interference, leading to unstable solution accuracy. In contrast, the proposed ELNCP-LCCZNN model maintains a consistent computational accuracy (around the order of 10^{-10}) across all three noise scenarios.

C. Application to Manipulator Control in Free of Noise

Application of the proposed ELNCP-LCCZNN to real-time kinematic trajectory tracking on a 6-DoF UR3 manipulator. Let $\theta \in \mathbb{R}^n$ be the joint-angle vector of the UR3 arm, and $\dot{\theta}$ be

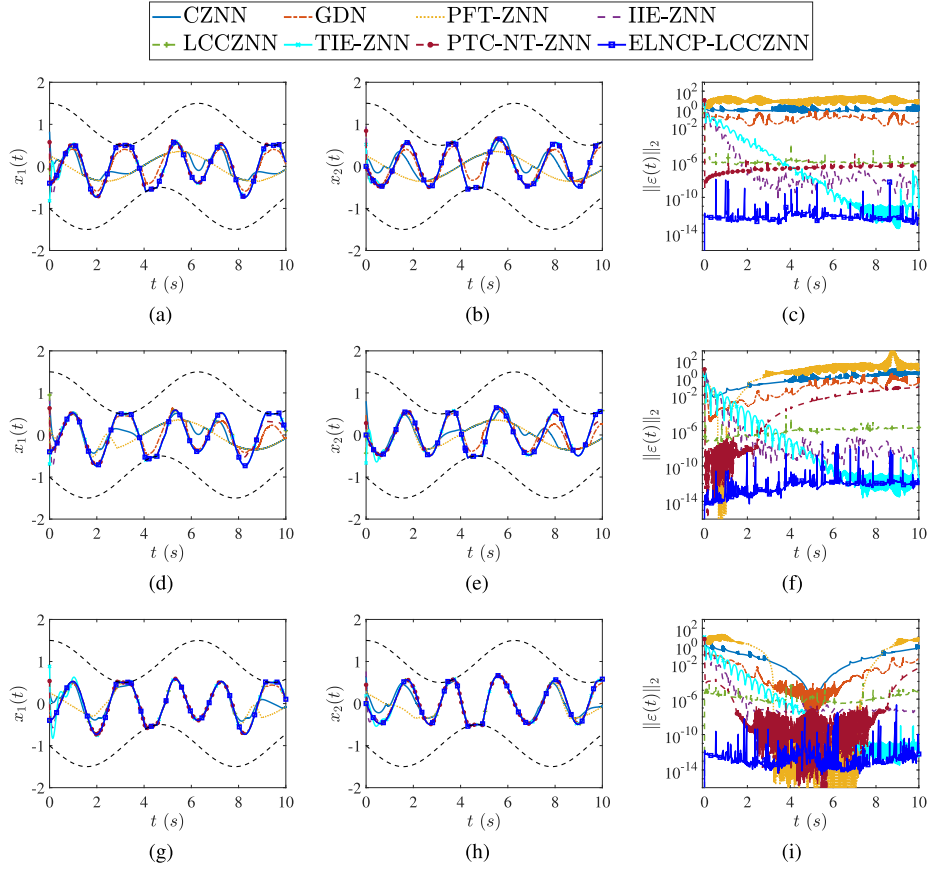


Fig. 3. Computational results obtained from existing RNN-based models, including CZNN, GDN, PFT-ZNN, IIE-ZNN, LCCZNN, TIE-ZNN, and PTC-NT-ZNN with time-varying noise when solving the TVQP problem (36). (a) Neural state x_1 generated by various RNNs with noise n_1 . (b) Neural state x_2 generated by various RNNs with noise n_1 . (c) Residual errors synthesized by various RNNs with noise n_1 . (d) Neural state x_1 generated by various RNNs with noise n_2 . (e) Neural state x_2 generated by various RNNs with noise n_2 . (f) Residual errors synthesized by various RNNs with noise n_2 . (g) Neural state x_1 generated by RNNs with noise n_3 . (h) Neural state x_2 generated by various RNNs with noise n_3 . (i) Residual errors synthesized by various RNNs with noise n_3 .

the corresponding joint-velocity vector. Based on this setting, a velocity-level motion planning problem with joint physical constraints can be formulated within a TVQP framework as

$$\text{minimize } \dot{\theta}^T(t)\dot{\theta}(t)/2 \quad (37)$$

$$\text{subject to } \mathbf{J}(t)\dot{\theta}(t) = \mathbf{r}_d(t) \quad (37a)$$

$$\dot{\theta}^- \leq \dot{\theta}(t) \leq \dot{\theta}^+ \quad (37b)$$

where $\mathbf{J}(t) \in \mathbb{R}^{m \times n}$ is the manipulator Jacobian relating joint velocities to task-space motion, while $\mathbf{r}_d$ is the desired end-effector velocity, and $\dot{\theta}^-$ and $\dot{\theta}^+$ specifies the lower and upper bounds of joint velocities, respectively.

According to the generalized TVQP formulation (1), the aforementioned motion control problem is formulated into a kinematics model, where the kinematic control variables and coefficient matrices are defined as follows:

$$\begin{cases} \mathbf{x}(t) = \dot{\theta}(t) \\ \mathbf{A}(t) = \mathbf{J}(\theta(t)), \mathbf{b}(t) = \mathbf{r}_d \\ \mathbf{Z}(t) = \mathbf{I}_6, \mathbf{u}(t) = -0.2, \mathbf{v}(t) = 0.2 \end{cases} \quad (38)$$

where upper $\mathbf{v}(t)$ and lower $\mathbf{u}(t)$ joint-velocity limits of each joint are set as -0.2 and 0.2 rad/s. In this validation scenario, UR3 robotic arm tracking of a standard circular path using the

TABLE IV

COMPARISON OF COMPUTER SIMULATION RESULTS AMONG CZNN [21], GDN [37], PFT-ZNN [6], IIE-ZNN [26], LCCZNN [35], TIE-ZNN [27], AND PTC-NT-ZNN [38] AND ELNCP-LCCZNN (26) IN CPU COST TIME (s), CONVERGENCE TIME (s), AND MEAN SQUARED ERROR (MSE) OF POSITION ERROR (m) FOR TRAJECTORY TRACKING CONTROL OF UR3 MANIPULATORS

Model	CPU Cost (s)	Convergence Time (s)	MSE (m)
CZNN [21]	1.1	7.97	3.69×10^{-3}
GDN [37]	13	2.38	4.86×10^{-4}
PFT-ZNN [6]	8.47	1.18	5.98×10^{-4}
IIE-ZNN [26]	2.66	12.74	6.93×10^{-4}
LCCZNN [35]	2.07	0.59	4.34×10^{-4}
TIE-ZNN [27]	2.87	13.06	6.65×10^{-4}
PTC-NT-ZNN [38]	10.26	0.97	3.14×10^{-4}
ELNCP-LCCZNN	2.38	0.08	1.29×10^{-4}

The bolded numbers indicate the smallest number in the column, making it easier to identify the best performance.

ELNCP-LCCZNN model in (26). In addition, the initial joint angles $\theta(0)$ are randomly generated within the interval $[0, 1]$ except for joint 2, where $\theta_2(0) \in [-1, 0]$. The initial values of the joint velocity and Lagrange multipliers are set to zero, and the model convergence parameter γ is set to 500.

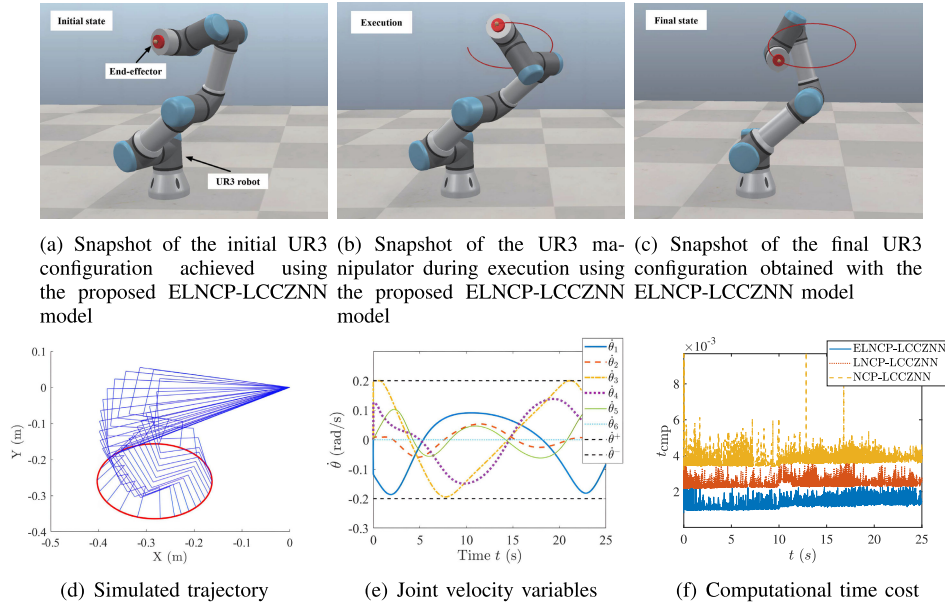


Fig. 4. Simulative results generated by the ELNCP-CZNN solvers when the simulated UR3 manipulators track a circle. (a) Simulated trajectory of the UR3 manipulators along a circle by LCCZNN. (b) Joint velocity variables of the UR3 manipulator along a circle. (c) Computational time of the LCCZNN with ELNCP, LNCP, and NCP. (d) Simulated trajectory. (e) Joint velocity variables. (f) Computational time cost.

The simulation is successfully conducted in the Virtual Robotics Experiment Platform (V-REP) integrated development environment, as illustrated in Fig. 4(a)–(c). The simulation results for UR3 robotic arm control using the NCPs-LCCZNN model in circular trajectory tracking are presented in Fig. 4(d)–(f). In this task, the joint velocities obtained from the NCPs-LCCZNN solver, as shown in Fig. 4(e), remain strictly within the constraint limits. Furthermore, the computational time costs analysis in Fig. 4(f) further validates the efficiency of the ELNCP framework, demonstrating that the ELNCP-LCCZNN solver achieves the shortest computational time cost. This enhanced efficiency improves the robotic arm’s responsiveness in practical applications. The aforementioned RNN-based models are also implemented under noise-free conditions to control the UR3 robotic arm for comparative analysis. The experimental results are presented in Table IV. The table summarizes the computational efficiency (CPU cost), convergence time, and position mean squared error (mse) achieved by each model during the circular trajectory tracking task with task duration T_{task} set to 25 s. The mathematical expression for the position mse is given as follows:

$$mse = \sum_{i=1}^N \left\| \mathbf{r}(t) - \mathbf{r}_d(t) \right\|_2 / N \quad (39)$$

where $N = T_{task}/\Delta t$, Δt denotes the sampling interval. Among these, CZNN exhibited the lowest CPU computational time consumption at 1.1 s, while the proposed ELNCP-LCCZNN model required 2.38 s of computation. This difference arises because STPEAF incorporates exponential and power operations, unlike traditional linear or polynomial AF (e.g., $\Phi(x) = x$ in CZNN), thereby introducing nonlinear computational overhead in each iteration. Increasing the STPEAF parameter reduces the theoretical convergence time but simultaneously

increases the computational cost of exponential and power operations. A reasonable parameter setting, typically within the range from 1 to 9, achieves a balance between convergence predictability and computational efficiency. Compared to other improved ZNN models that also employ nonlinear AFs, the proposed model achieves substantially lower computational cost. Furthermore, under comparable parameter settings, the proposed model outperforms other models in both convergence time and position mse, achieving a convergence time of 0.08 s and a position mse of 1.29×10^{-4} .

D. Application to Manipulator Control Under Noise

The theoretical analysis and numerical simulations presented above demonstrate the superior robustness of the ELNCP-LCCZNN model against noise. The proposed ELNCP-LCCZNN model is evaluated under linear time-varying noise conditions defined as $\mathbf{n}_2 = 0.5t$, for a UR3 robotic arm trajectory tracking task to validate its robustness. Specifically, the ELNCP-LCCZNN model (27) under noise is employed to control the UR3 robotic arm to track a circular trajectory, with its performance compared with that of the aforementioned RNN-based models under identical noisy conditions. The corresponding results are presented in Fig. 5. Fig. 5(a)–(p) displays the actual trajectories generated by each model and quantifies the corresponding position errors. Simulation results show that, under noisy conditions, the CZNN, GDN, and PFT-ZNN models exhibit progressive deviations from the desired trajectory, with positional errors accumulating over time. In contrast, despite the presence of noise, the remaining models maintain precise trajectory-tracking performance, completing tracking tasks. A comparison of their position error plots reveals that the proposed ELNCP-LCCZNN model maintains tracking accuracy on the order of 10^{-4} under noise interference, outperforming the order of 10^{-3}

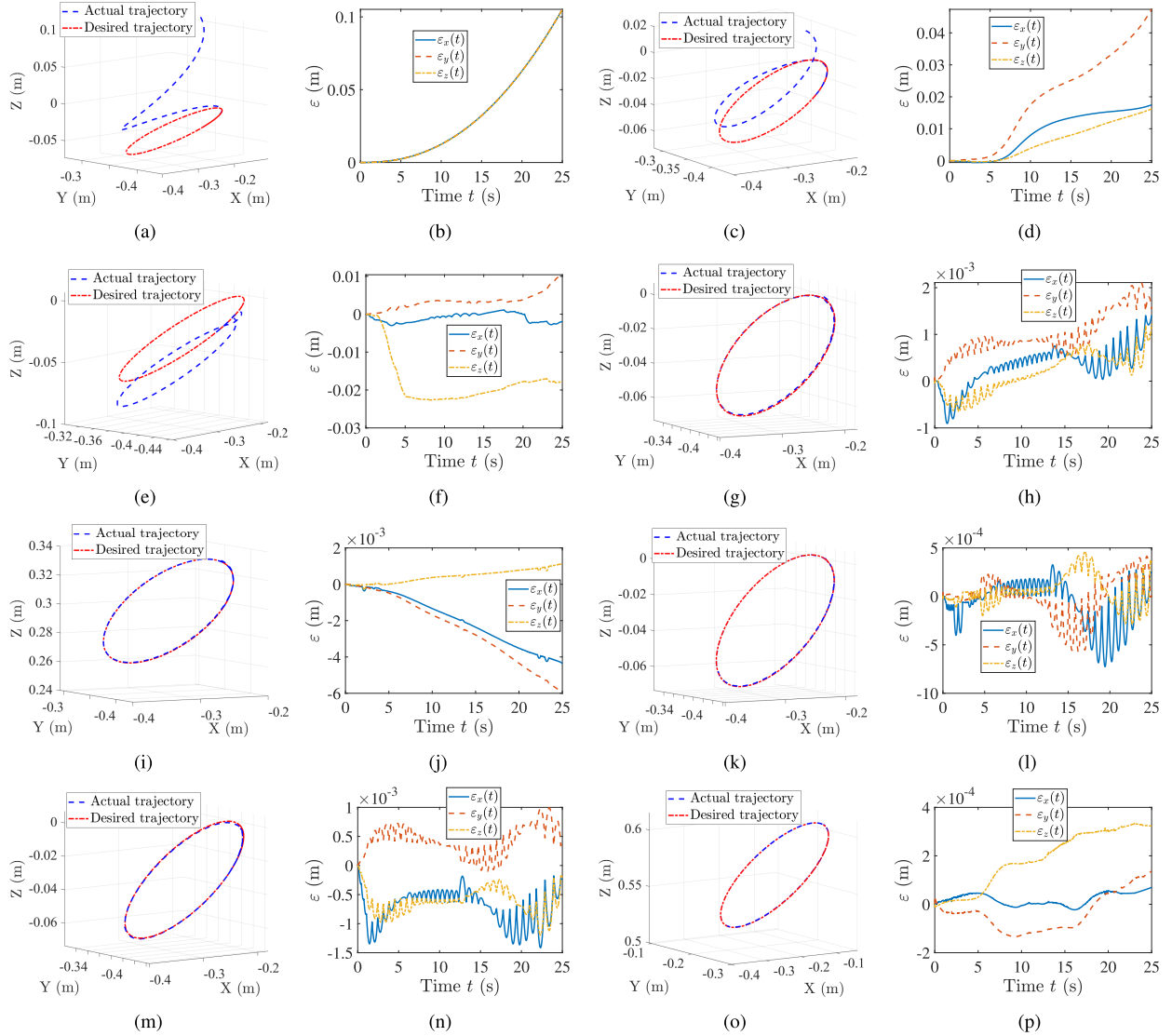


Fig. 5. Comparison of actual and desired trajectories synthesized by the UR3 robotic manipulator using aforementioned RNN-based models under linear time-varying noise $n_2(t) = 0.5t$, along with corresponding trajectory errors. (a) Trajectories generated by CZNN. (b) Trajectory errors synthesized by CZNN. (c) Trajectories generated by GDN. (d) Trajectory errors synthesized by GDN. (e) Trajectories generated by PFT-ZNN. (f) Trajectory errors synthesized by PFT-ZNN. (g) Trajectories generated by IIE-ZNN. (h) Trajectory errors synthesized by IIE-ZNN. (i) Trajectories generated by LCCZNN. (j) Trajectory errors synthesized by LCCZNN. (k) Trajectories generated by TIEZNN. (l) Trajectory errors synthesized by TIE-ZNN. (m) Trajectories generated by PTC-NT-ZNN. (n) Trajectory errors synthesized by PTC-NT-ZNN. (o) Trajectories generated by ELNCP-LCCZNN. (p) Trajectory errors synthesized by ELNCP-LCCZNN.

accuracy achieved by other models. The experimental results further confirm that the ELNCP-LCCZNN model exhibits superior robustness to noise interference, making it highly suitable for robotic applications requiring stable and precise motion control.

E. Experimental Validation With UR3(e) Robot

This section validates the engineering applicability of the ELNCP-LCCZNN model through physical experiments conducted on a UR3(e) robotic manipulator. Specifically, the localization task requires continuous tracking of a circular trajectory with a predefined radius of $r = 8$ cm, while the orientation task ensures that the Z-axis of the end-effector coordinate system remains perpendicular to the operational plane, as illustrated in Fig. 6. This type of task is

relevant to industrial applications such as precision welding. The robotic system is initialized with joint angles $\theta(0) = [0; -\pi/2; \pi/2; 0; \pi/2; 0]$ rad, while joint velocity $\dot{\theta}(0)$ and Lagrange multipliers λ_1 and λ_2 are initialized as zero vectors. The tracking performance is presented in Fig. 6(a)–(c), where the Z-axis of the end-effector remains perpendicular to the drawing board throughout the experiment, and the actual trajectory closely aligns with the desired trajectory. A direct comparison between the desired and actual trajectories is provided in Fig. 6(d). In addition, Fig. 6(e) and (f) depicts the joint velocity and trajectory position error results of the end-effector, respectively. The results demonstrate that the joint velocity remains within the prescribed velocity constraints, while the trajectory position error remains within 1.5×10^{-4} m, consistent with the simulation results from Section V-C. In summary, these experimental findings provide compelling evi-

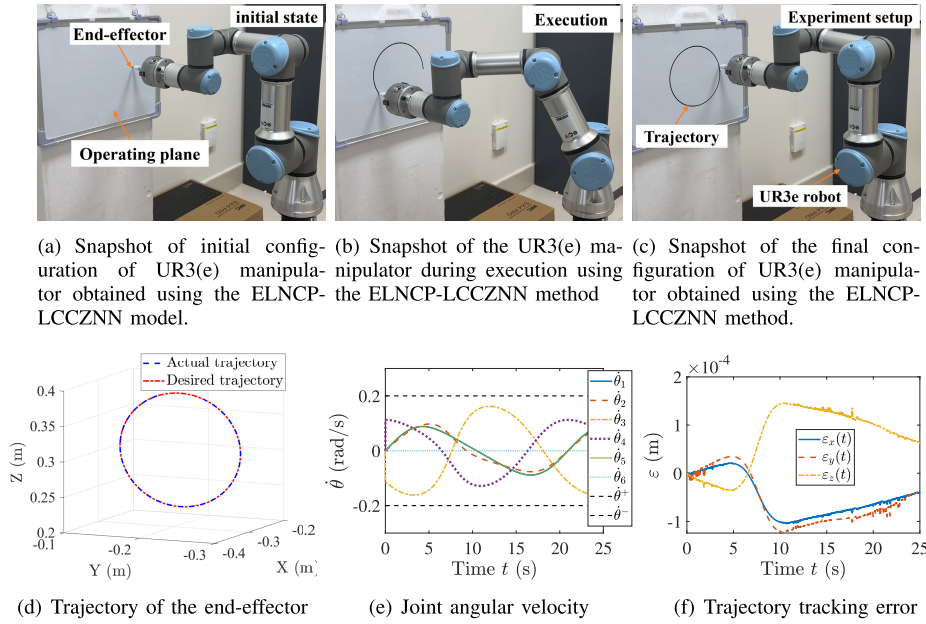


Fig. 6. Experimental results on a UR3(e) manipulator tracing a circular path synthesized by the ELNCP-LCCZNN. (a)–(c) Photograph of robot states at different times. (d) Trajectory of UR3(e). (e) Joint angular velocity of UR3(e). (f) Position error of the end-effector.

dence supporting the effectiveness and accuracy of the position control scheme described in (37).

VI. CONCLUSION

This article proposes an ELNCP-LCCZNN model equipped with the STPEAF to address bounded constrained TVQP problems. By integrating a novel reduced-dimension NCP function into the LCCZNN framework, the proposed approach further decreases computational complexity while preserving convergence and robustness properties. Numerical simulations demonstrate that the proposed method significantly reduces the physical solution time compared with existing boundary constraint handling approaches, resulting in a faster system response. Furthermore, comprehensive comparisons with state-of-the-art RNN methods demonstrate that the proposed framework achieves superior overall performance in terms of convergence speed, solution accuracy, and robustness against noise. Simulation and experimental results on the UR3 and UR3e robot platforms validate the capability of the proposed approach to provide rapid response and strong noise rejection, as well as its feasibility and stability in practical applications. Future work will extend this framework to higher dimensional systems, more complex constraint conditions, and multiobjective optimization scenarios.

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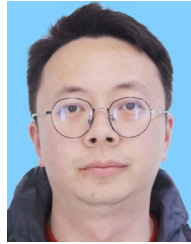
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